duration missions. The ground control station serves as the operational hub, managing mission planning and data interpretation. Collectively, the figure emphasizes how the integration of communication, computation, and control defines third-generation drones' versatility and technological sophistication.

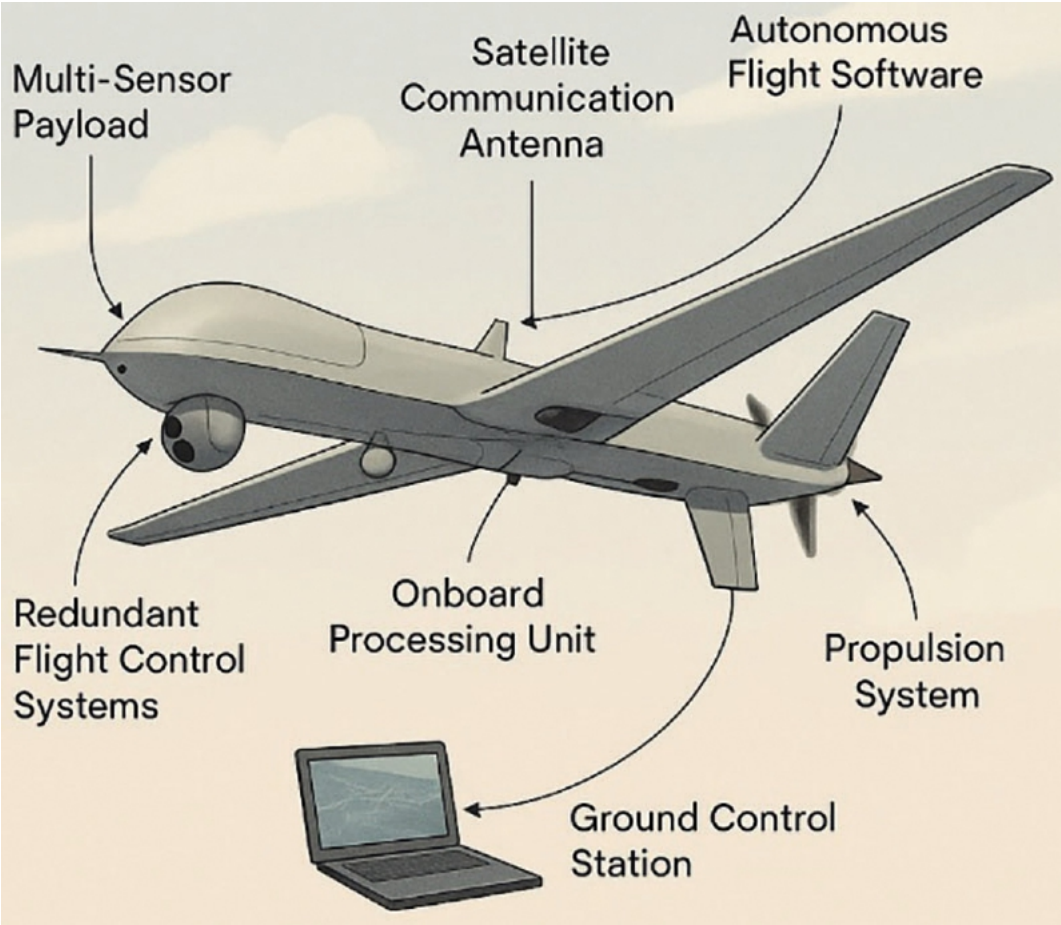


Figure 4.1 Structural and functional layout of third-generation UAVs.

4.3.1 Principles of operation

- *Autonomous mission management*: Third-generation UAVs are built to execute end-to-end mission profiles, take off, transit, surveillance/strike, and recovery with minimal human intervention. Mission plans (waypoints, sensor tasks, weapons employment logic)